

Golf Genius (GG) <-> Smartscore (SS) Integration Flow

Document Purpose: Step-by-step integration flow summary for GG-SS real-time score synchronization

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1. Overview

1.1 Integration Goal

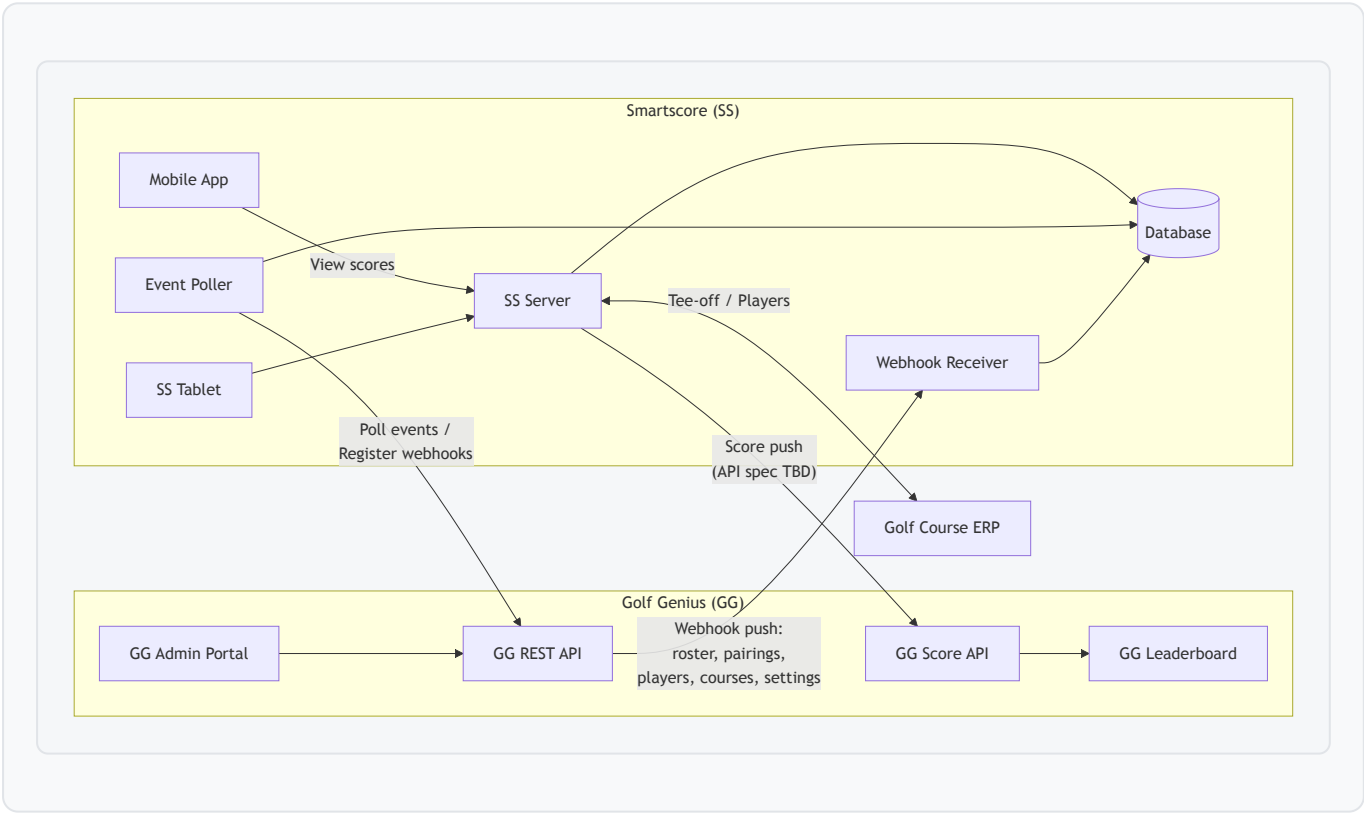
The primary goal of this integration is to **push real-time scores from SS tablets to GG** so that GG can provide tournament leaderboards, standings, and result services for events managed through GG.

1.2 Data Flow Direction

Direction	Data	Method
GG -> SS (Inbound)	Events, Roster, Pairings, Courses, Settings	Polling + Webhooks (Hybrid)
SS -> GG (Outbound)	Real-time scores	GG Score API (exact spec TBD by GG)

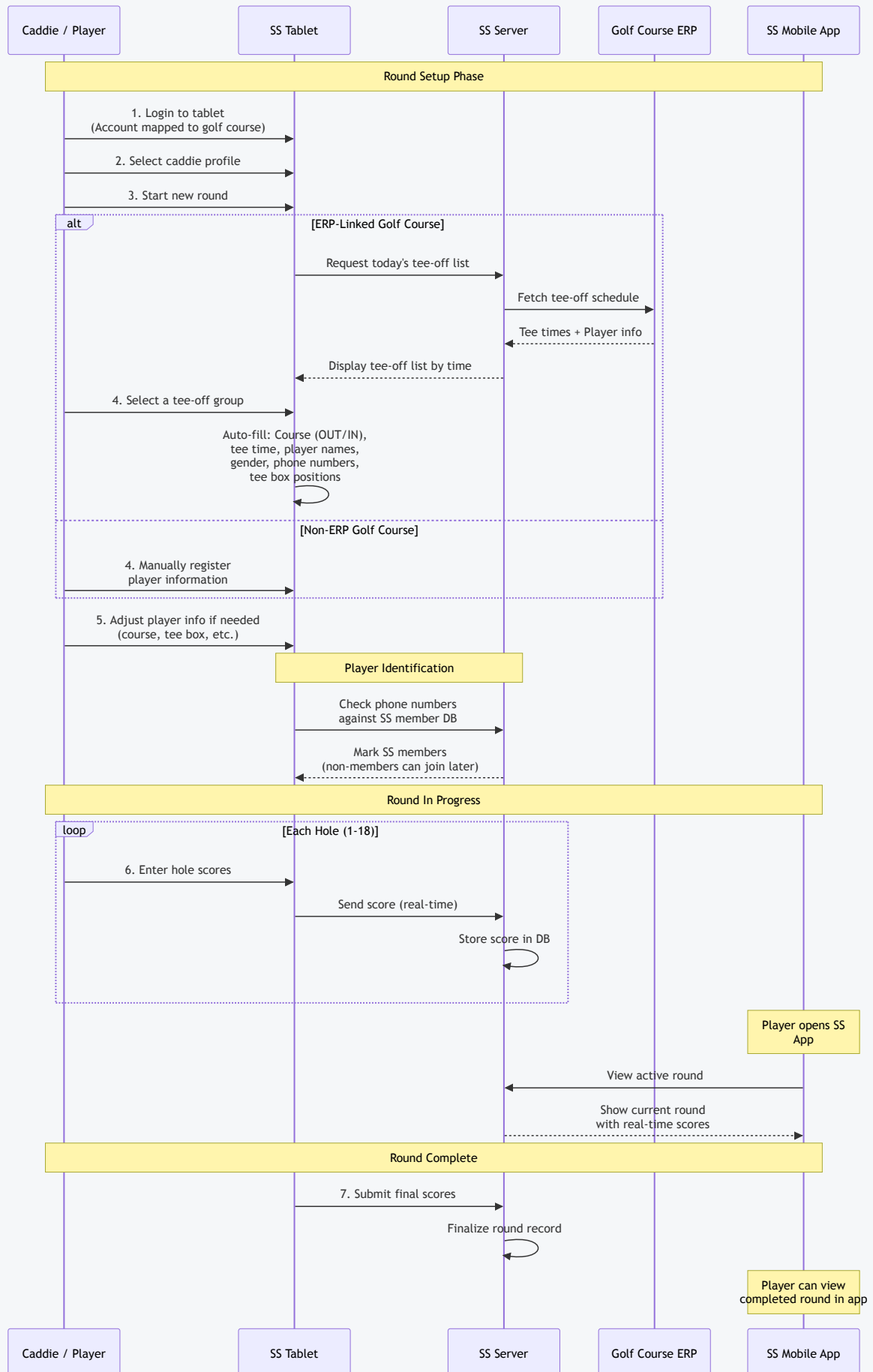
2. System Context

2.1 High-Level Architecture



2.2 SS Tablet Round Flow (Existing)

To understand where the GG integration fits, here is the existing SS tablet round flow:



Key Points:

- The tablet account is pre-mapped to a specific golf course
- If the course has ERP integration, tee-off schedules are automatically loaded
- Players are identified by **phone number** (primary identifier in SS)
- Scores are transmitted in real-time, hole by hole
- Players can open the SS mobile app to see their active round and check current scores (pull-based, not push notification)
- Non-SS-members who provide a phone number can later register and claim their scores within a certain period

3. End-to-End Integration Flow

3.1 Complete Integration Lifecycle

Phase 1: Initial Setup

GG issues API Key
per golf course



SS Admin stores API Key
in SS course config



SS enables GG sync
for the course



Phase 2: Event Discovery

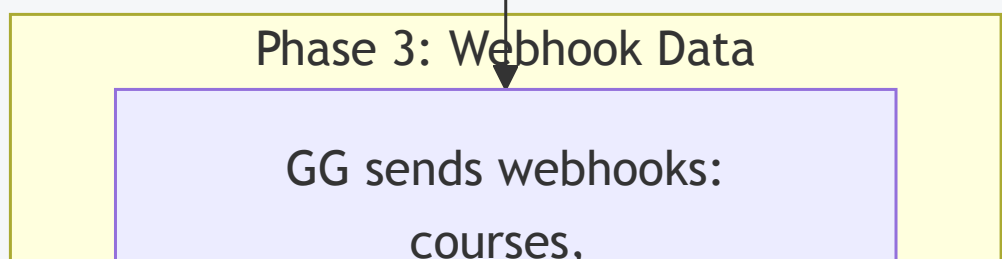
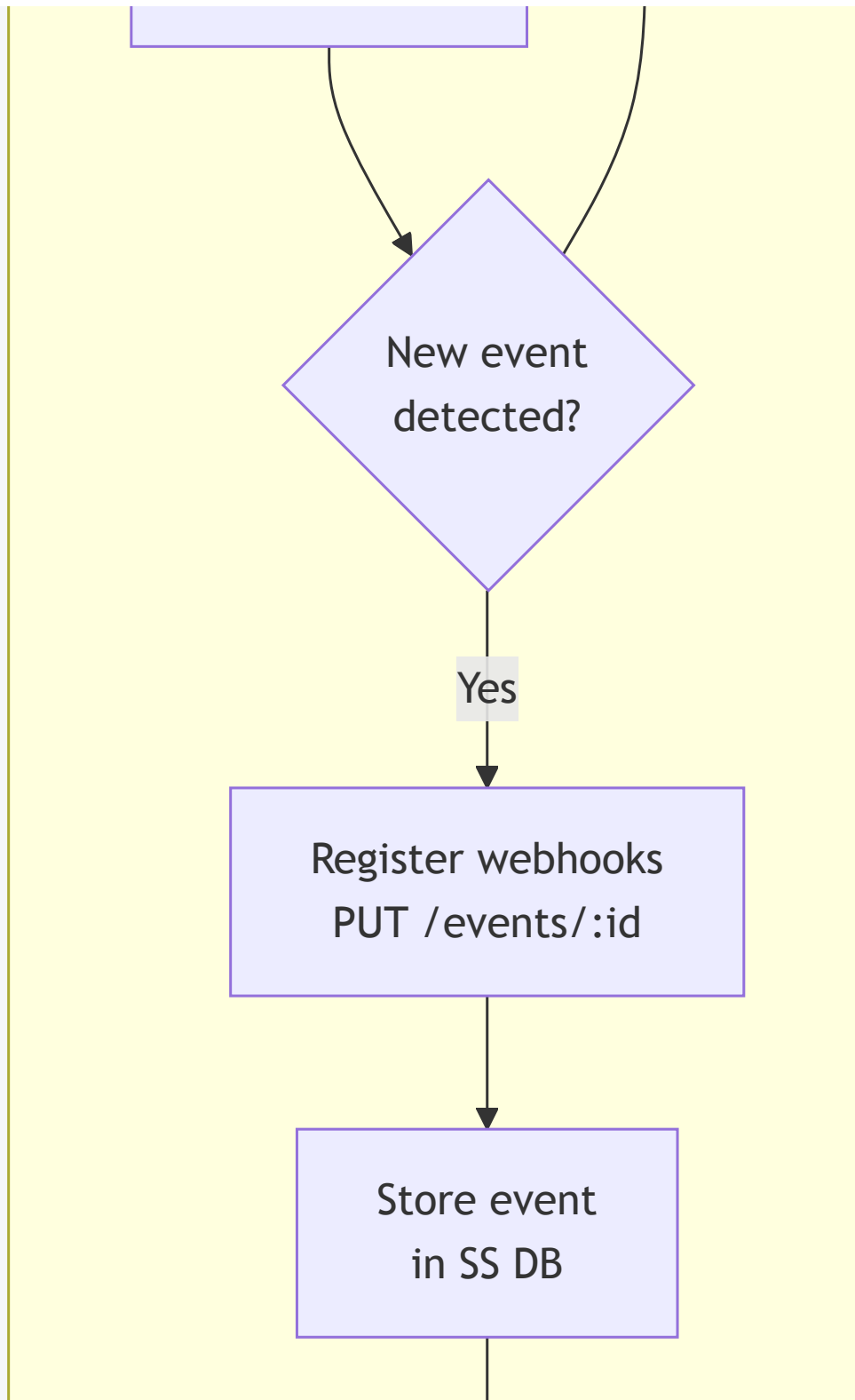
SS Poller runs
every 10 min



GET /events
from GG API

No





event_roster_members,
pairings, players, settings



SS stores all data
in local DB



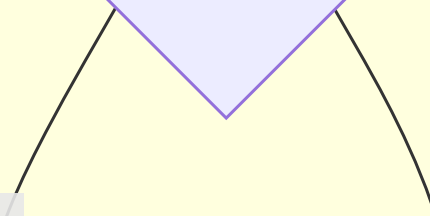
Phase 4: Tablet Round

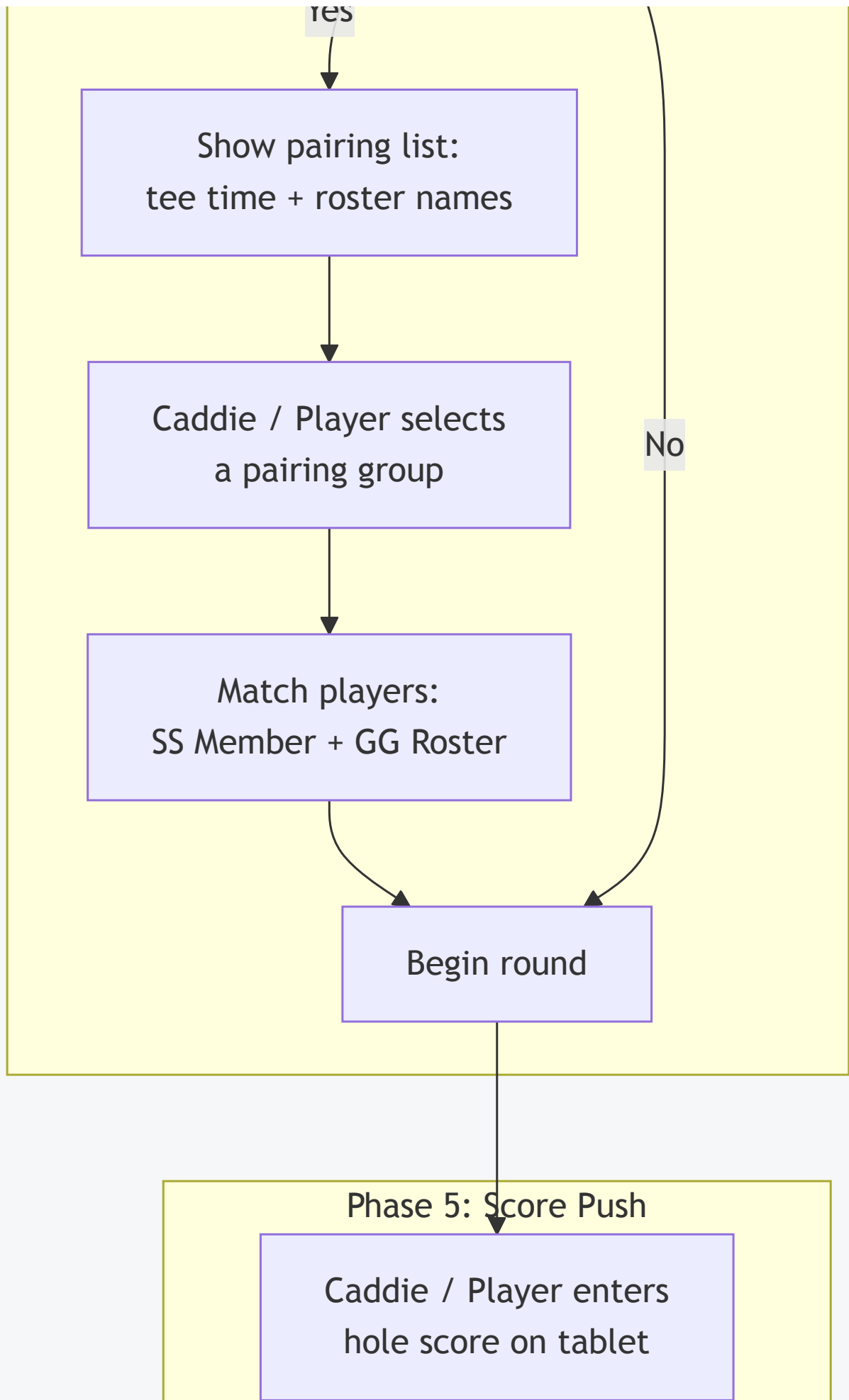


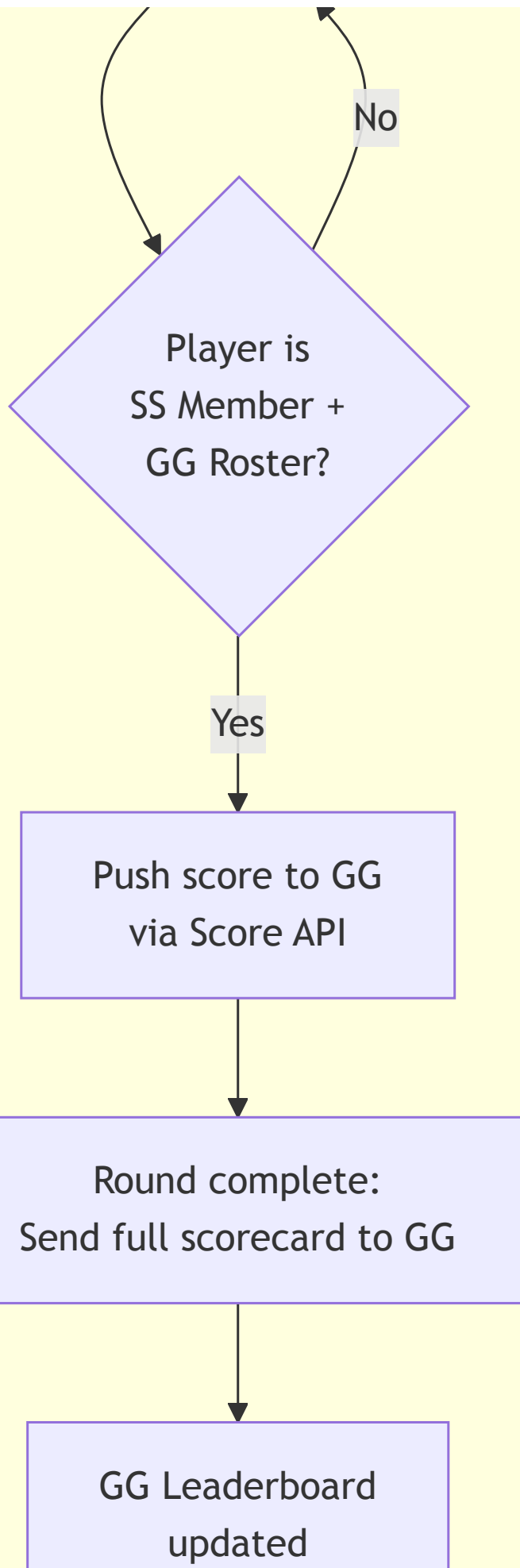
Caddie / Player starts
round on tablet



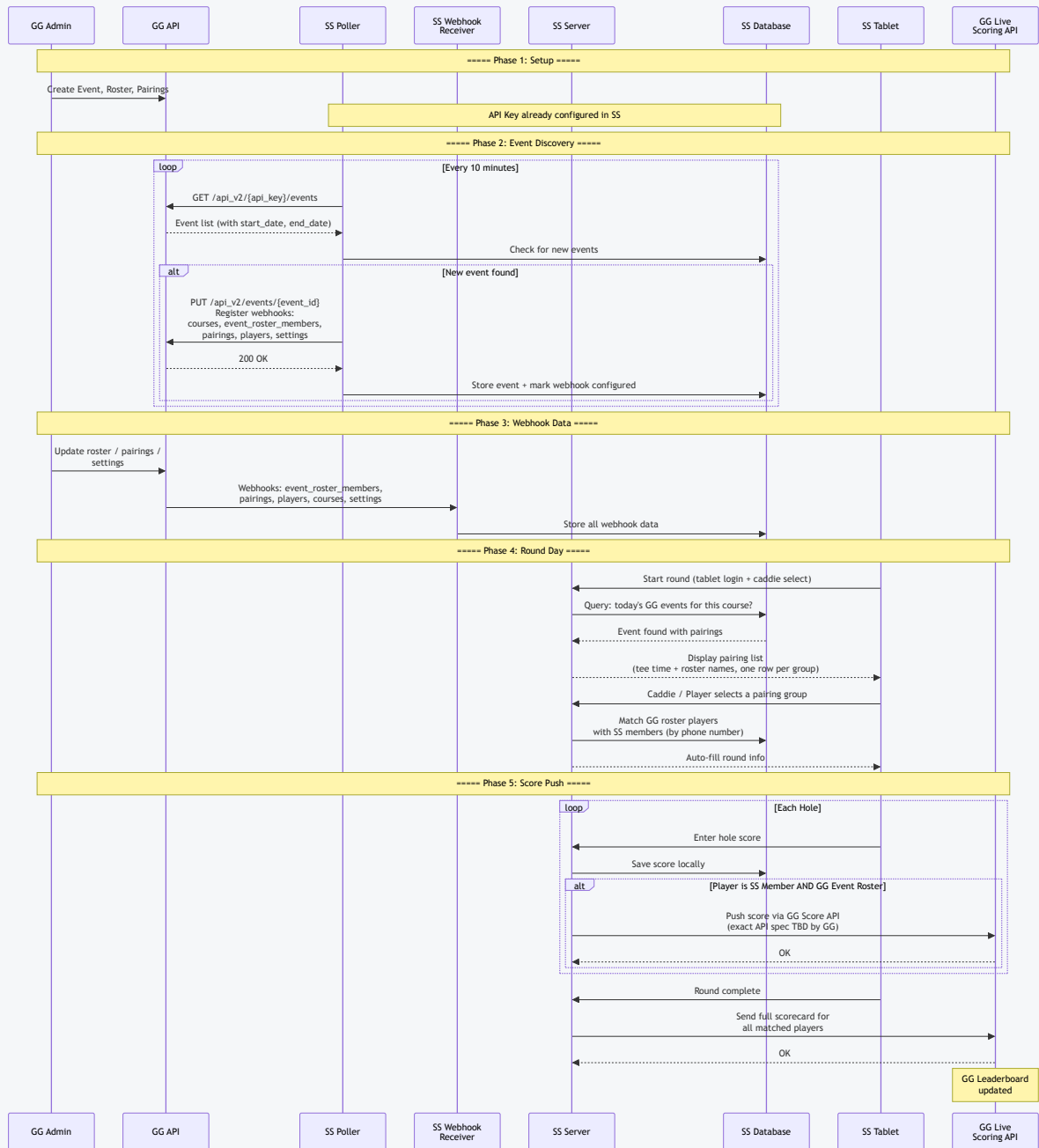
Today's GG event
exists?







3.2 Full Sequence Diagram

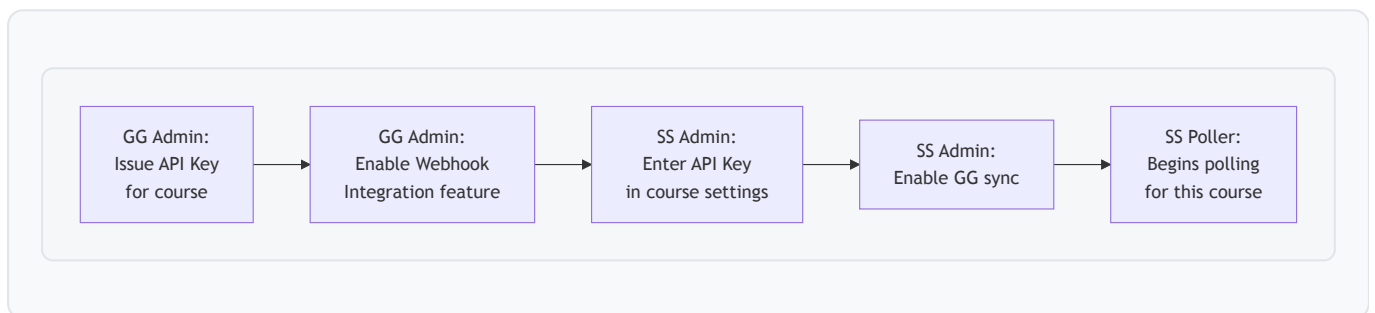


4. Phase 1: Initial Setup & Configuration

4.1 Prerequisites

Item	Responsible	Description
GG API Key	GG	Issued per golf course (Customer Center)
Webhook Integration Feature	GG	Must be enabled per customer (Admin Center > Edit > Product Versions)
SS Course Configuration	SS	Store API Key in SS database, enable GG sync

4.2 Configuration Flow



4.3 SS Database Configuration

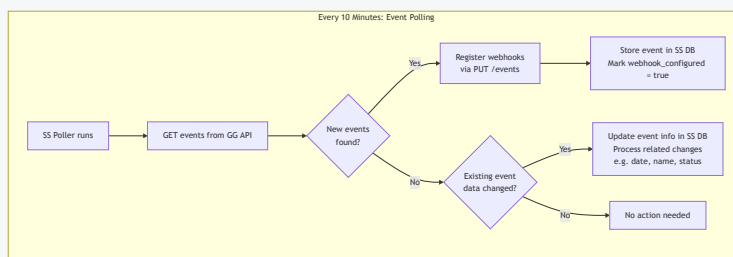
SS stores the following per golf course:

Field	Description
<code>api_key</code>	GG API Key for this course
<code>webhook_types</code>	Which webhooks to register (default: courses, event_roster_members, pairings, players, settings)
<code>sync_enabled</code>	Whether GG sync is active
<code>score_push_enabled</code>	Whether to push scores to GG

5. Phase 2: Event Discovery & Webhook Registration

5.1 Event Polling Strategy

SS polls GG for new events at a **10-minute interval**. When a new event is detected, SS automatically registers webhooks for that event via the GG API.



5.2 Webhook Types

Webhook	Purpose	Default
<code>courses</code>	Course layout changes (holes, pars, tees)	Enabled
<code>event_roster_members</code>	Players added/removed from event roster	Enabled
<code>pairings</code>	Pairing group changes (groups, tee times, player assignments)	Enabled
<code>settings</code>	Event/round setting changes	Enabled
<code>players</code>	Individual player profile changes (name, handicap)	Enabled
<code>scores</code>	Score changes (SS->GG direction, so disabled by default)	Optional
<code>matches</code>	Match play information	Optional
<code>match_results</code>	Match play results	Optional
<code>team_results</code>	Team competition results	Optional
<code>teams</code>	Team assignments	Optional

5.3 Webhook Re-registration Strategy

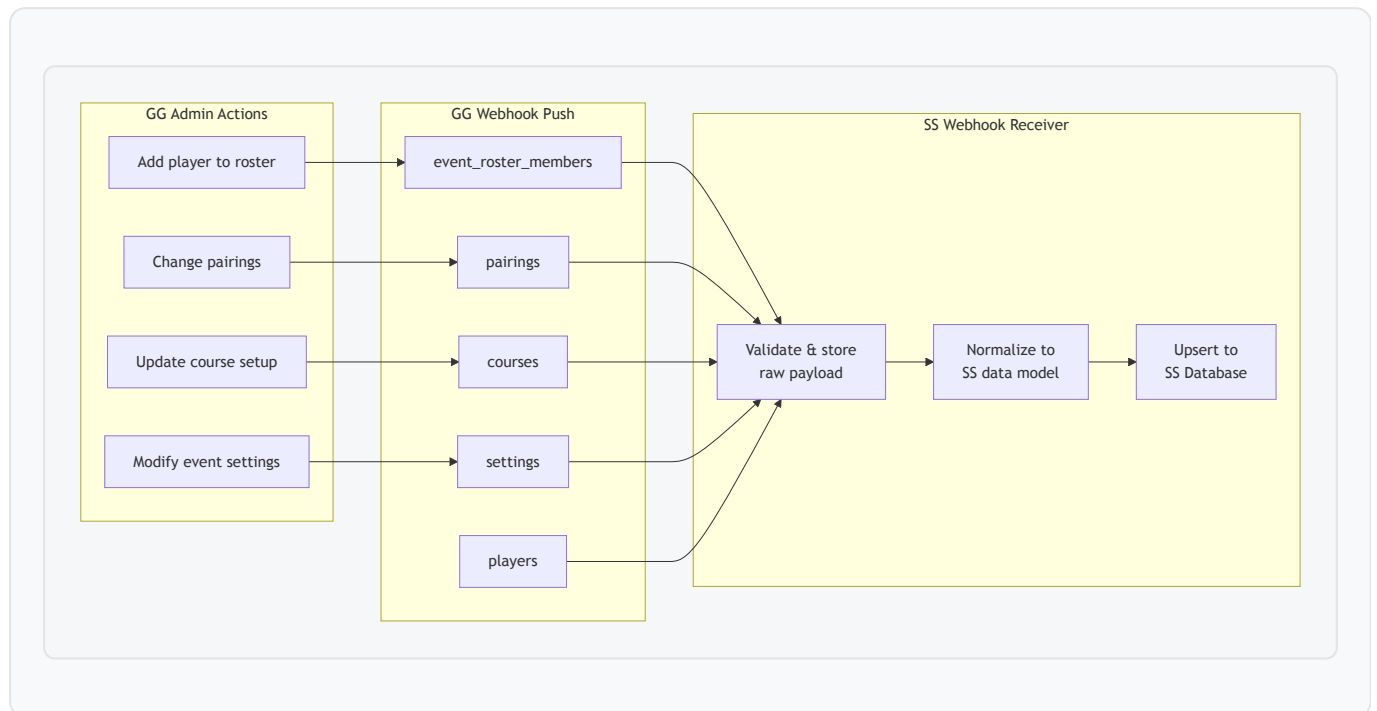
Trigger	Interval	Target
New event detected	On detection (via 10-min poll)	New events only
Active event refresh	Every 1 hour	Events with today's rounds
Full re-sync	Once per day	All non-archived events

Why re-register? If a GG Admin manually modifies webhook settings through the GG Admin Portal, SS re-registration ensures webhook endpoints are always correctly configured.

6. Phase 3: Receiving Master Data via Webhooks

6.1 Webhook Data Flow

Once webhooks are registered, GG pushes data to SS whenever changes occur in the event.



6.2 Key Data Received via Webhooks

Event Roster Members

Contains the list of players registered for the event:

Field	Description
<code>member_id</code>	GG Master Roster member ID
<code>first_name</code> / <code>last_name</code>	Player name
<code>email</code>	Player email (GG primary identifier)
<code>handicap_index</code>	Player handicap
<code>division_id</code>	Division assignment
<code>team_id</code>	Team assignment (if applicable)
<code>custom_fields</code>	May include phone number

Pairings

Contains pairing groups (foursomes) with tee times:

Field	Description
<code>pairing_group_id</code>	Unique group ID
<code>tee_time</code>	Starting time (e.g., "08:00 AM")
<code>starting_hole</code>	Starting hole number (for shotgun starts)
<code>players[]</code>	Array of players in this group
<code>players[].roster_entry_id</code>	Player's roster entry ID (needed for score push)
<code>players[].player_round_id</code>	Player's round-specific ID

Courses

Contains course layout information:

Field	Description
<code>course_id</code>	Course ID
<code>holes[]</code>	Hole definitions (par, yardage, handicap index)
<code>tees[]</code>	Tee box definitions

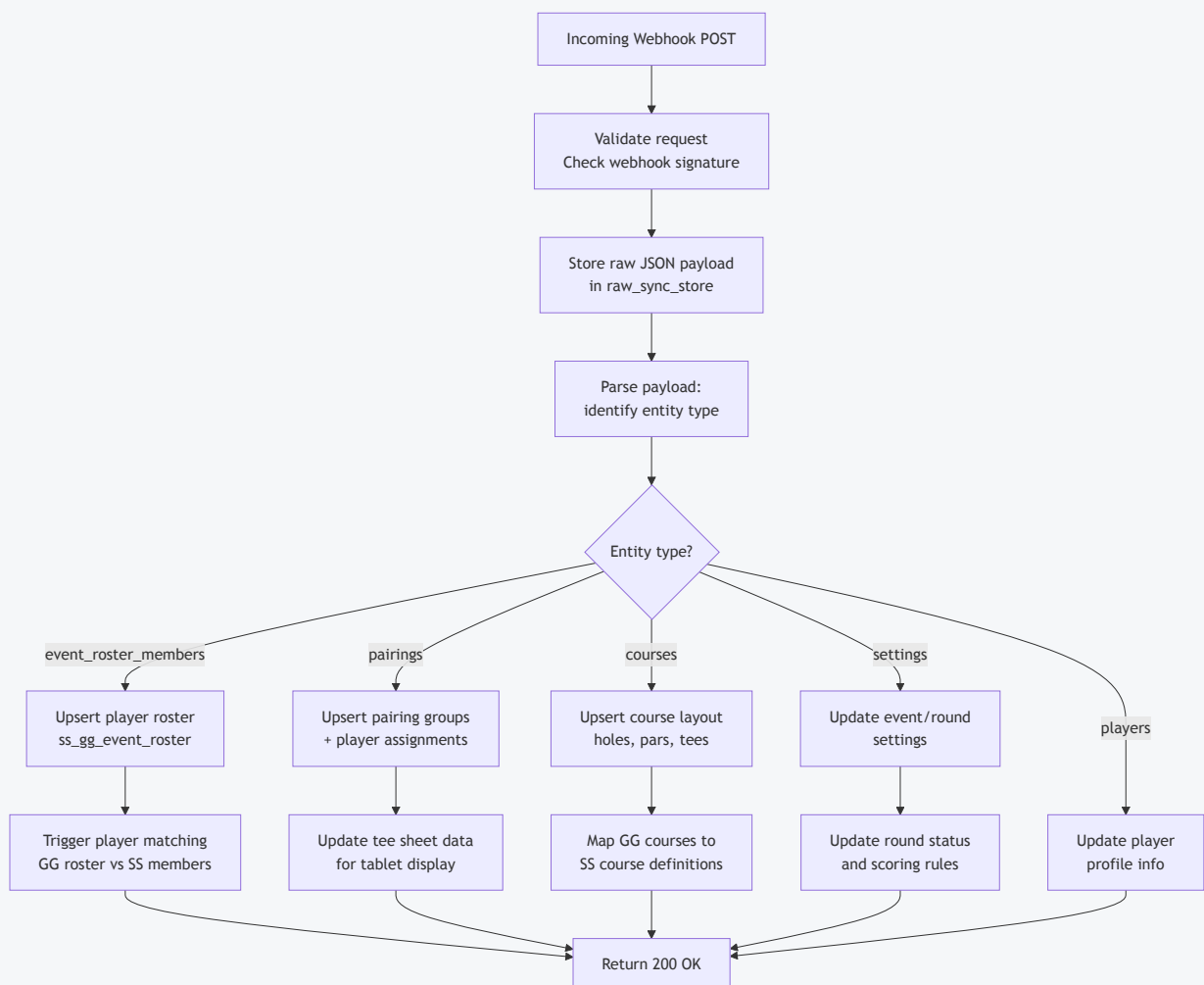
6.3 SS Webhook Endpoints

All webhooks are received at:

Base URL: <https://golf-genius.smartscore.kr/ss/gg/webhooks/>

Endpoint	Method	Purpose
<code>/ss/gg/webhooks/courses</code>	POST	Course data changes
<code>/ss/gg/webhooks/event_roster_members</code>	POST	Roster member changes
<code>/ss/gg/webhooks/pairings</code>	POST	Pairing/tee-time changes
<code>/ss/gg/webhooks/settings</code>	POST	Event/round setting changes
<code>/ss/gg/webhooks/players</code>	POST	Player profile changes

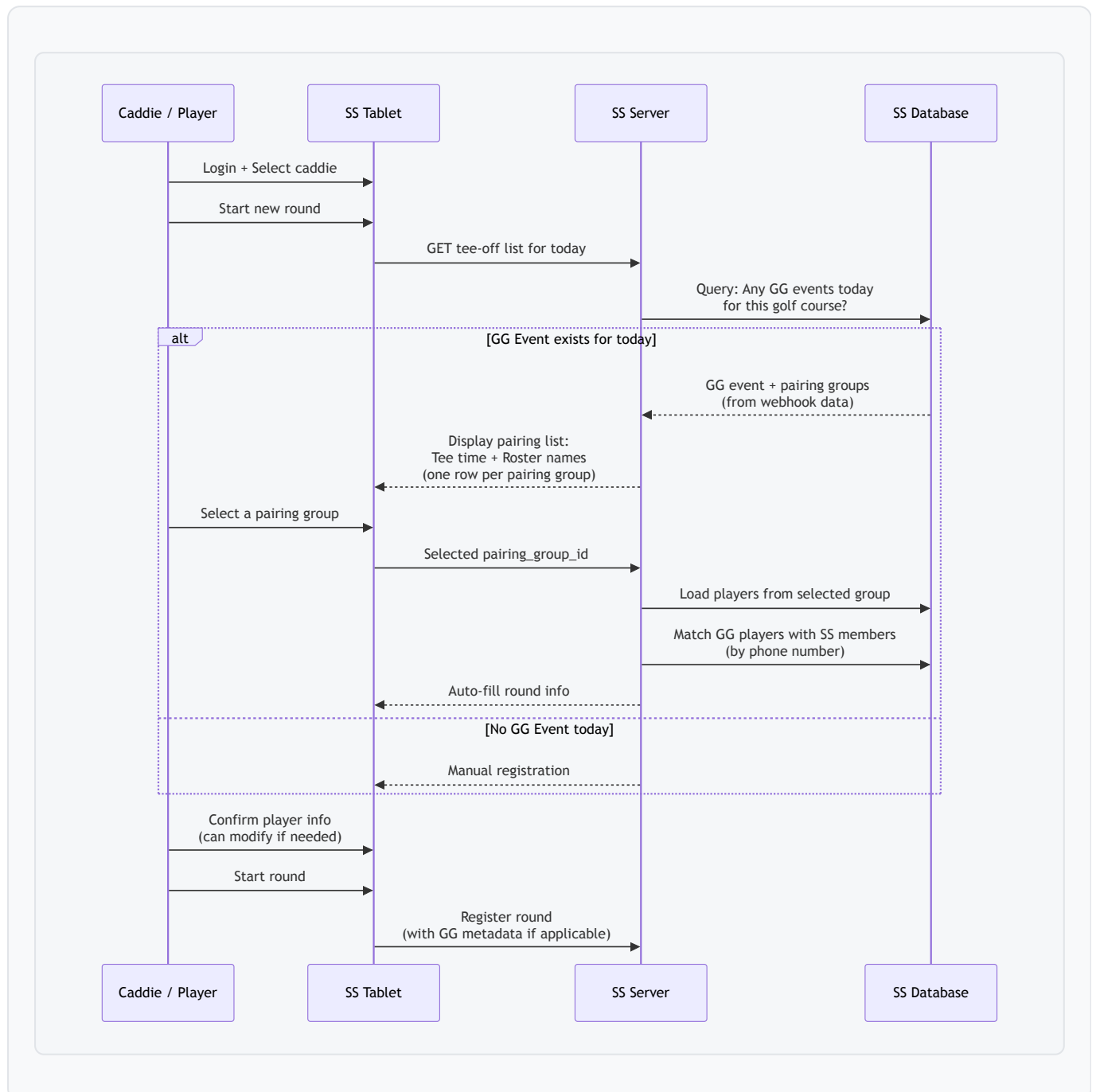
6.4 Webhook Processing Pipeline



7. Phase 4: Tablet Round Flow with GG Integration

7.1 Enhanced Tablet Flow

When a caddie or player starts a round on the tablet, if a GG event exists for today, the pairing data received via webhooks is presented as a selectable list.



7.2 Data Source on Tablet

Note: Whether the tablet displays the existing ERP tee-off list or the GG Event Roster list (or both) is **to be determined**. This decision will be made at a later stage.

The data displayed on the tablet for GG events is:

- Tee-off time
- Roster player names

Each pairing group is shown as **one row** in the list.

7.3 Tablet Display: GG Pairing List

When GG event data is available, the tablet shows a simple list:

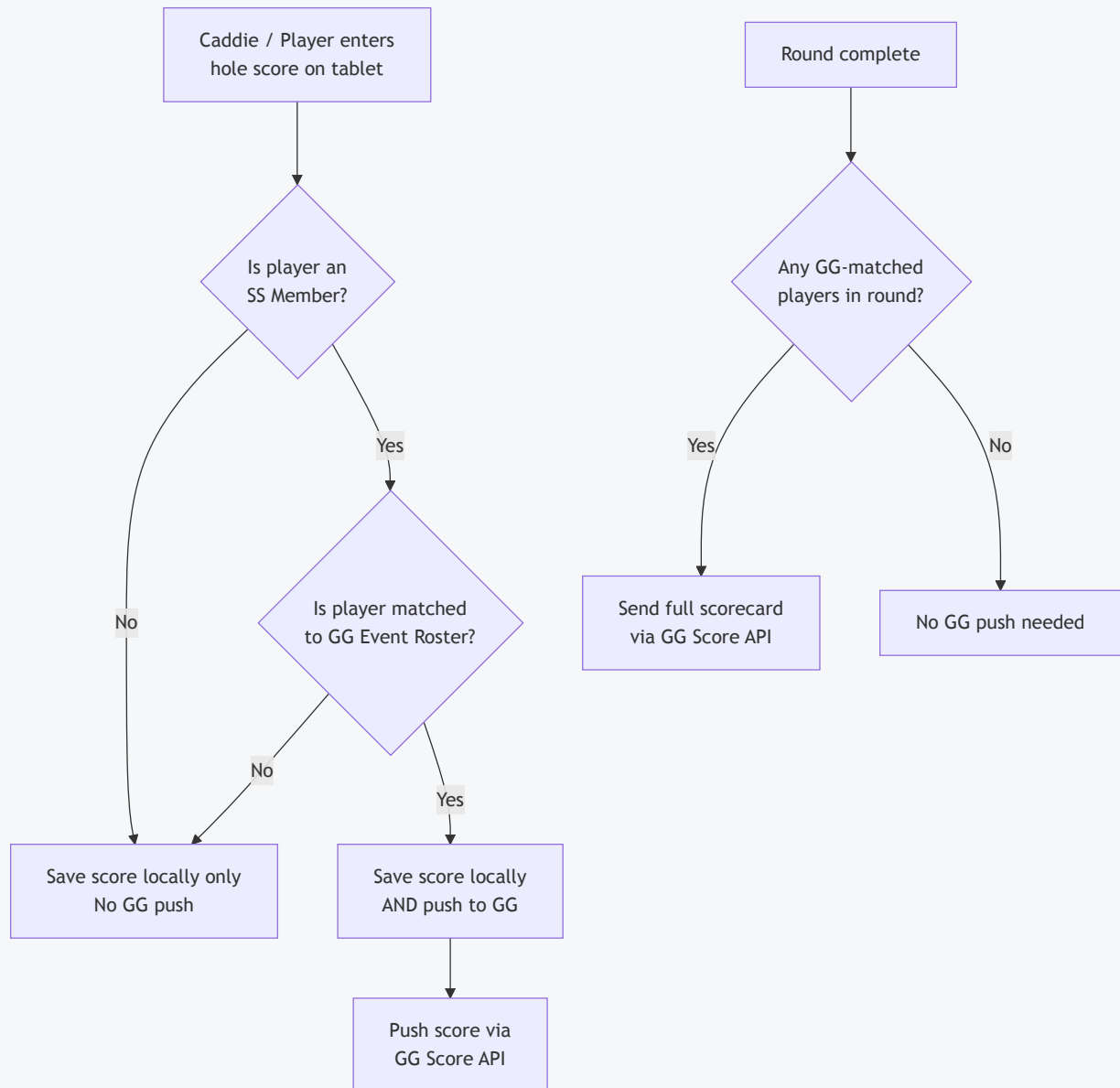
08:00		Kim Cheolsu, Lee Younghee, Park Minsu, Choi Jihoon
08:10		Jung Sumin, Kang Daeun, Yoon Seojun, Han Jiwoo
08:20		Hong Gildong, Seo Jiyeon, Cho Minho, Baek Dahye
...		

8. Phase 5: Real-Time Score Push (SS to GG)

8.1 Score Push Eligibility

Scores are pushed to GG only when **both conditions** are met:

- The player is an **SS Member** (identified by phone number in SS)
- The player is matched to a **GG Event Roster entry** (matched by phone number)



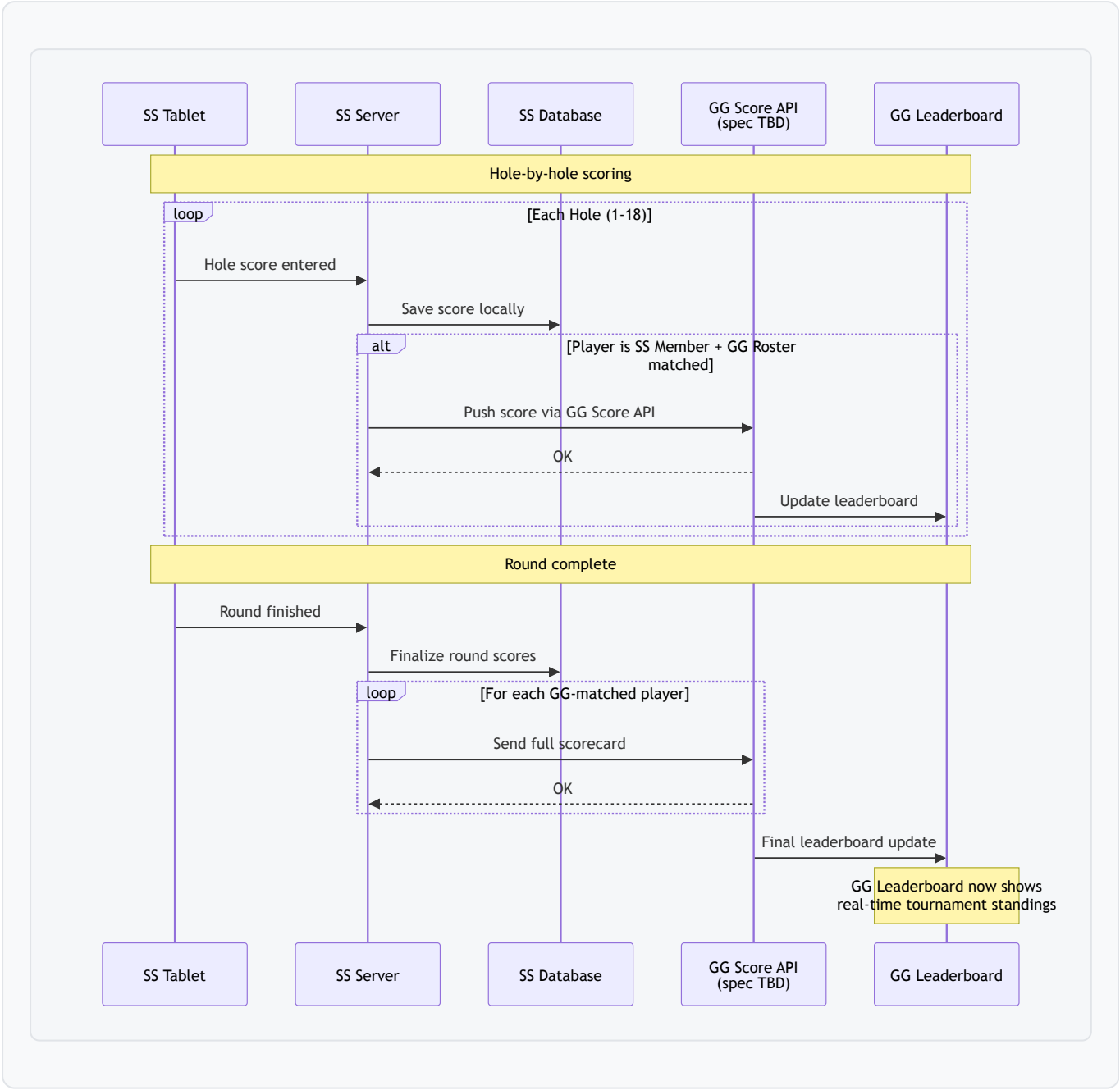
8.2 Score Push API

Note: The exact Score API specification (endpoints, request/response format, required identifiers) is **to be provided by GG**. Once GG confirms the final API spec, SS will implement the score push accordingly.

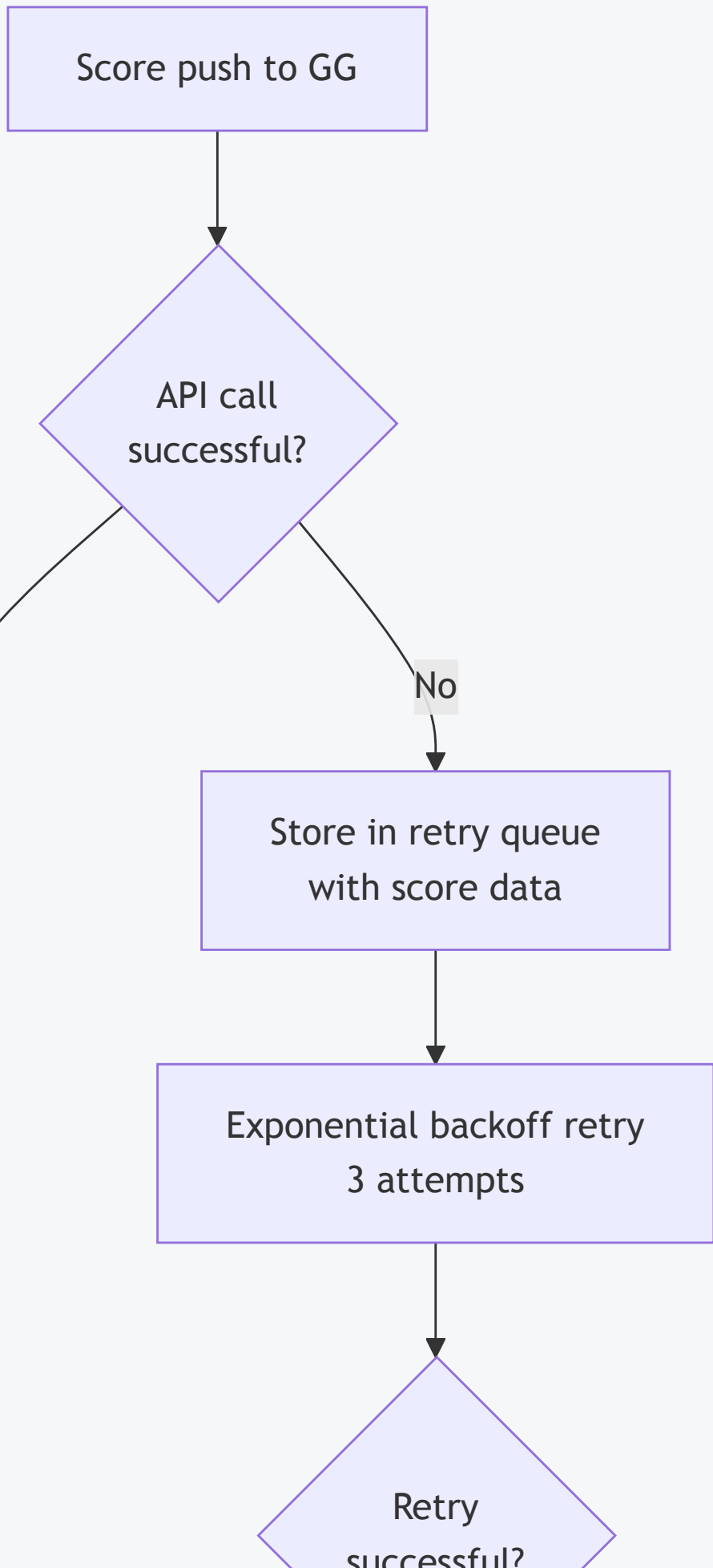
Expected capabilities:

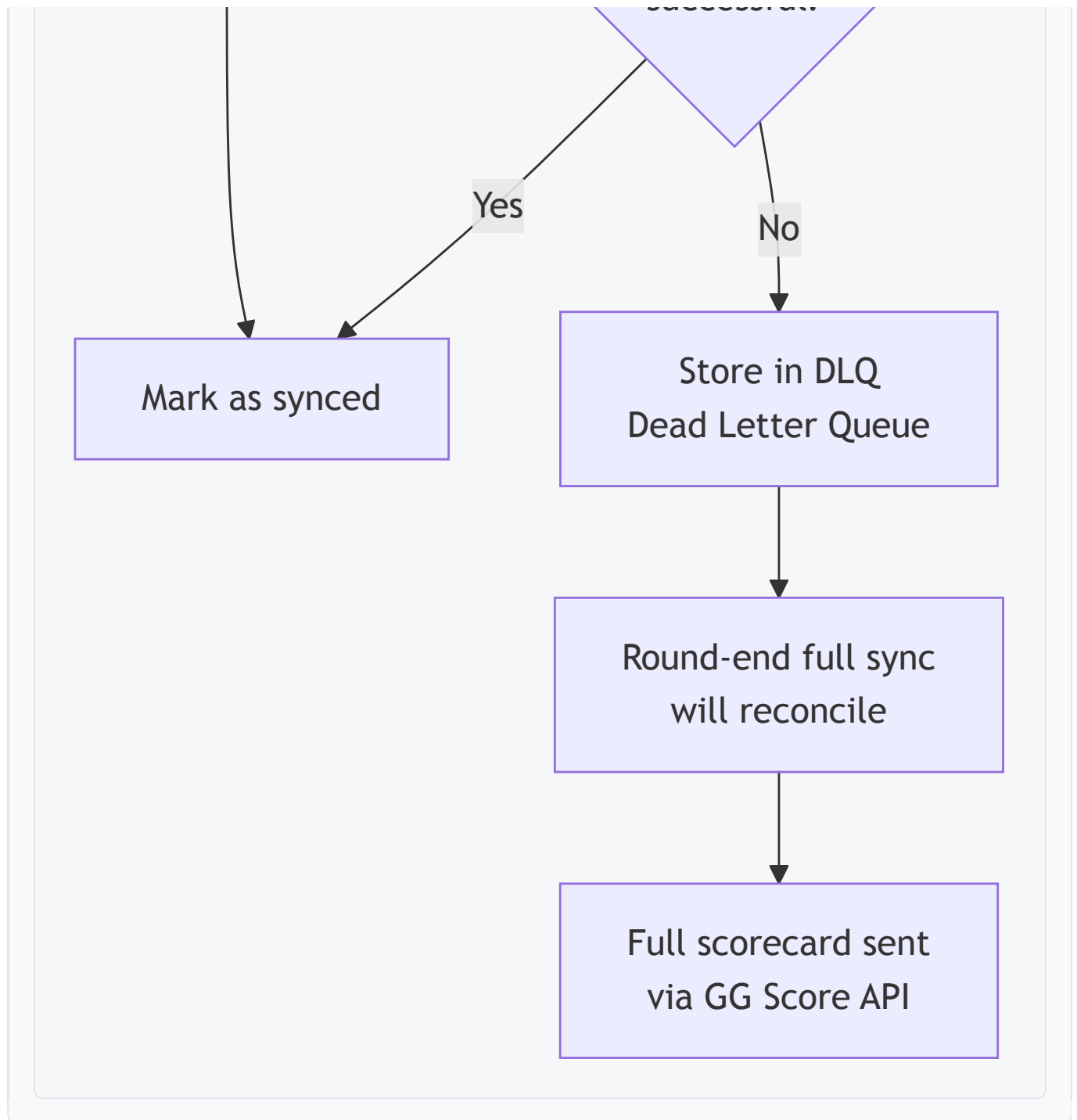
- Real-time score push (hole-by-hole or batch) during a round
- Full scorecard submission on round completion
- Required identifiers: event ID, round ID, player/roster entry ID, hole number, strokes, etc.

8.3 Score Push Flow Diagram



8.4 Network Failure Handling for Score Push





9. Player Matching Strategy

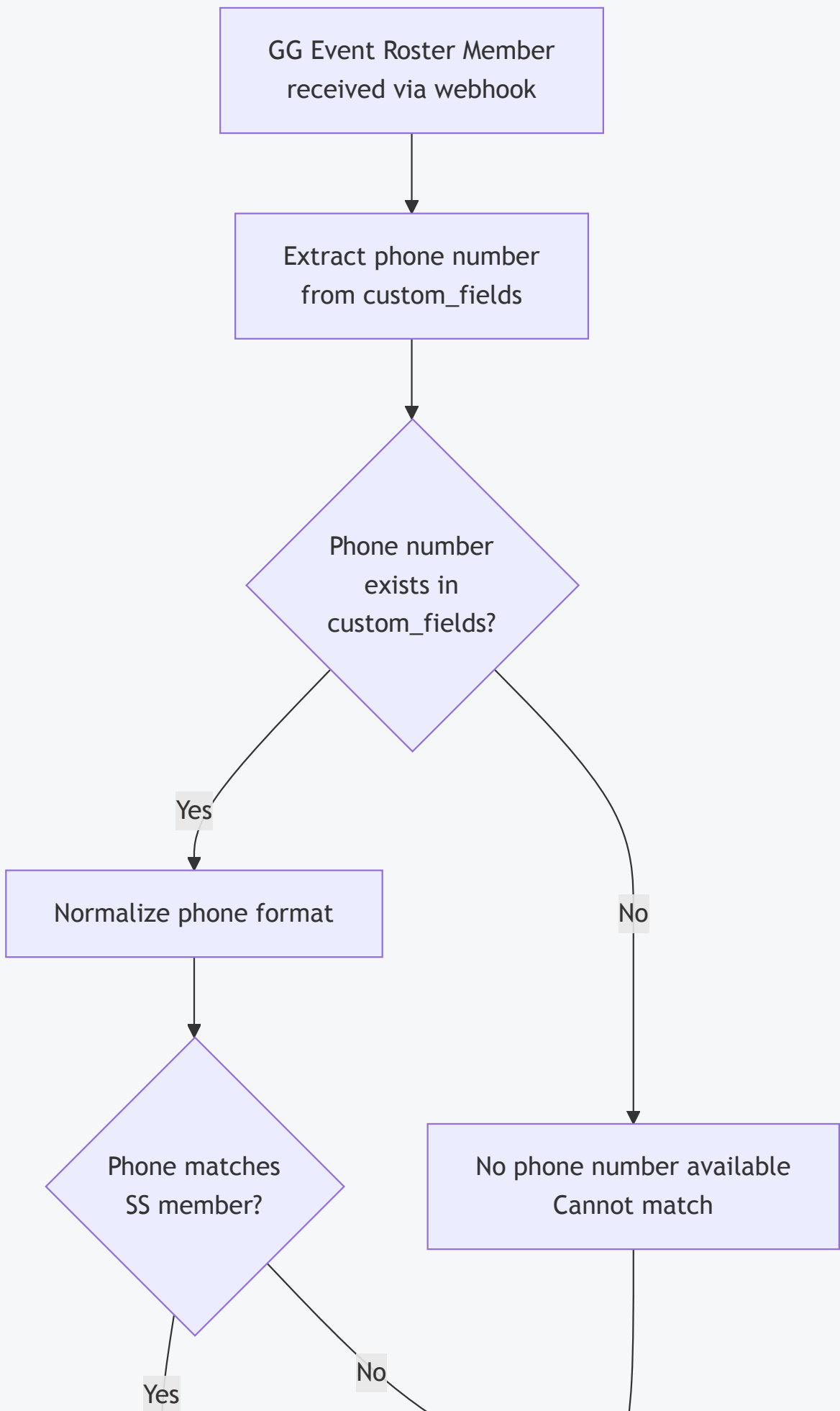
9.1 The Matching Challenge

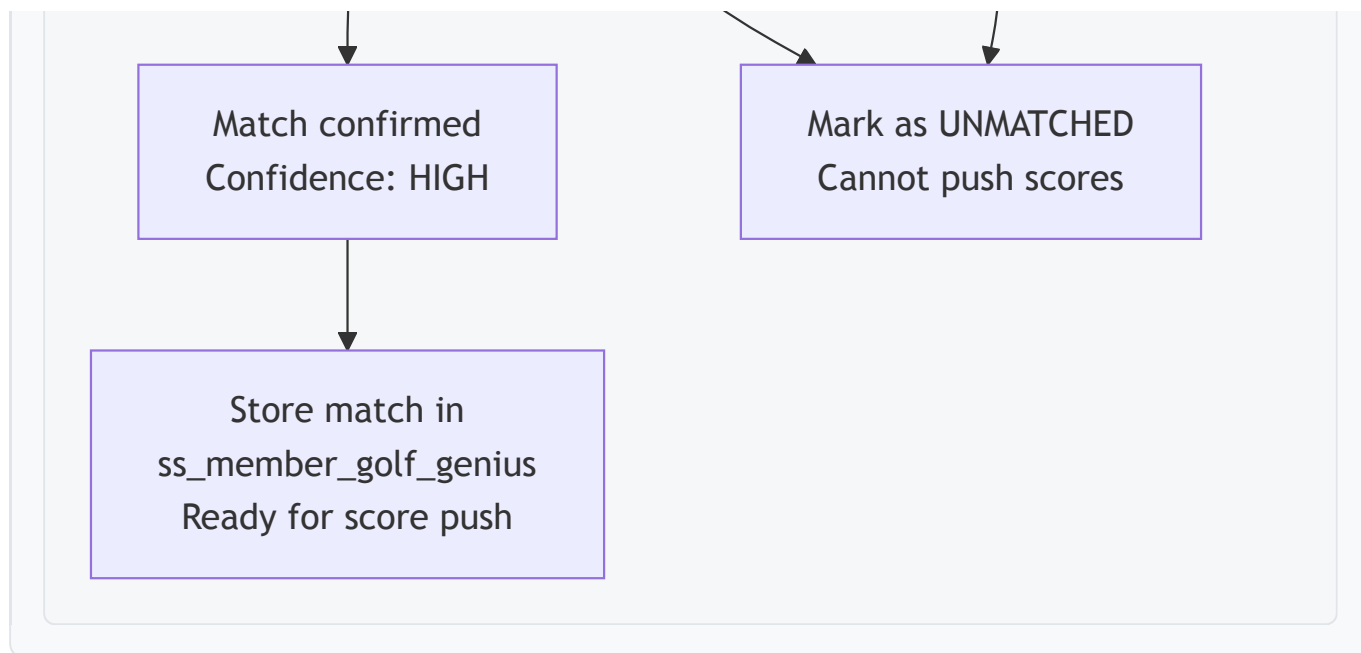
GG and SS use **different primary identifiers**:

System	Primary Identifier	Notes
GG	Email	Phone number is not a default field; can only be added via <code>custom_fields</code>
SS	Phone number	SS authenticates members via phone number verification. Email is not currently supported for matching.

This difference is a **critical integration challenge**. SS relies exclusively on phone numbers for member identification, but GG does not have phone number as a standard field.

9.2 Matching Flow





9.3 Match Priority

Priority	Method	Confidence	Auto-match?
1st	Phone number (from <code>custom_fields</code>)	High	Yes
-	No phone number available	-	Cannot match; scores not pushed to GG

9.4 Identifier Notes

- SS authenticates all members via **phone number verification** at sign-up
- SS tablets identify players by **phone number** during round registration
- GG does not include phone number as a default player field; it is only available via `custom_fields`
- Phone format normalization is required before matching (e.g., `010-1234-5678` vs `01012345678`)
- Once matched, the mapping is stored persistently and reused across events
- **This matching strategy depends on GG providing a reliable, standardized phone number field** (see Open Questions)

10. Open Questions for GG

The following items require clarification from the GG team to finalize the integration design:

Webhooks

#	Question	Context
1	What is GG's webhook retry policy on delivery failure?	To understand the overlap with SS polling
2	Does the <code>players</code> webhook fire only for event-scoped players or all master roster changes?	Determines whether SS needs separate master roster polling

Score API

#	Question	Context
3	What is the complete Score API specification (endpoints, request/response format)?	SS needs the final spec to implement score push

Player Matching (Phone Number)

#	Question	Context
4	SS authenticates members via phone number verification and uses phone numbers to identify players on tablets during rounds. However, GG's default player profile does not include a phone number field — it can only be added via <code>custom_fields</code> . When registering players to the event roster, is there a way to make phone number a required field with a standardized field name (e.g., <code>mobile_no</code>)? Since <code>custom_fields</code> allows free-form naming, different courses could use different field names (e.g., "Phone Number", "Cell Phone", "Mobile"), making it impossible for SS to reliably extract the phone number programmatically.	Phone number is the only identifier SS can use for player matching. Without a consistent, required phone number field, SS cannot match GG roster members to SS members and therefore cannot push scores to GG.

Course Mapping

#	Question	Context
5	On the SS tablet, players select a Front 9 course (OUT), Back 9 course (IN), and optionally an Additional 9-hole course when starting a round. SS manages courses as individual 9-hole units. For example, if a golf course has 27 holes with three 9-hole courses named "South", "East", and "West", SS needs to map the GG Course ID for each course to the corresponding SS internal course ID. SS plans to store the Course ID generated by GG upon course registration and map it to the SS course ID. This mapping is necessary because when pairings webhook data is received, SS needs to determine which SS course corresponds to the GG course in the pairing group, so the tablet can correctly auto-fill the OUT/IN course selection. Is the Course ID from GG's course registration the correct identifier to use for this mapping?	SS manages courses as 9-hole units (OUT/IN/Additional). Each golf course typically has 2-3 nine-hole courses. The pairings webhook includes course information, and SS must map it to internal course IDs so the tablet can auto-fill the correct course when a pairing group is selected.